

Institute of Actuaries of India

Subject SA3 – General Insurance

October 2014 Examination

INDICATIVE SOLUTION

Solution 1:

- (i) Per kilometer rating will be one measure of exposure for the damage portion of comprehensive motor, but is not the only relevant measure.
It will not be relevant for theft component, is relevant for collision, property damage and bodily injury. There still needs to be other rating factors such as postcode, age of driver, make of car etc

[2]

- (ii) • Refining a risk classification will not, everything else being equal, change the aggregate amount of claims incurred by an insurance company. There is no reason to expect that insurers' aggregate premium income will increase as a result of risk classifications becoming more refined and accurate.
- It will merely change the allocation of premium charges amongst its policyholders. If people who pose a higher risk are charged more when the sources of risk can be better identified in advance, then those who pose a lower risk will be certainly be charged less.
 - Insurers pursue refinements in risk classification procedures to reduce cross subsidizations between groups of insureds
 - If a competitor introduces a new rating factor and the company does not adopt it then the competitor will be able to attract better risks, leaving the less risk refined insurers with the riskier policyholders at premium levels which were set assuming an incorrect blend of risk categories.

[4]

- (iii) Alternative approaches to estimating mileage:

Motorists would prepay for the kilometers they expect to drive in a year and then pay more or receive a refund at the end of the year depending on actual kilometers driven.

Advantages:

- simple and free to collect

Disadvantages

- only able to audit at time of claim
- since only auditing at time of claim this means only part of the exposure data is audited
- if over kilometers estimated and claim denied, negative publicity

Use electronic devices that automatically record and transmit mileage data. Most new cars electronically record mileage in the engine computer, and new technologies such as global positioning system (GPS) transponders provide a means of wireless transmission.

Advantages:

- accurately reflects mileage
- no action required from consumer

Disadvantages

- cost – both of providing unit to policyholder and receiving information
- privacy concerns if also collecting location information

Annual declaration of vehicle usage and driving record

Advantages:

- simple and free to collect

Disadvantages

- only able to audit at time of claim
- if over kilometers estimated and claim denied, negative publicity

[4]

(iv) The first step in determining an appropriate set of relativities is to get an understanding of the underlying differences in technical cost by number of kilometers driven. The challenge would be limited data available.

- Would be unable to use GLM due to lack of data
- As kilometer data is not available, could use other rating factors to approximate distance driven. One example would be to use external information such as geographical codes, vehicle usage (do you drive to work).

Another example would be to use the existing rating data available, such as socio-economic conditions, location, make of car and driver age to provide an indicative 'score' for each kilometer driven band;

- Using some sort of credibility approach to determine the additional impact of kilometers driven band not accounted for by the factors identified above – i.e. for driving bands with sufficient exposure we might rely on purely their own claims experience but for other driving bands we would also consider the experience of other bands.
- A key thing to consider would be interactions between kilometer travelled and other rating variables.

Once we have determined the differences in kilometer driven bands, the next step is to overlay the usual pricing considerations such as where we choose to deviate from the technical cost to account for expected demand etc. To assist in these decisions we could model quote conversion and retention. We could also supplement this analysis with a better understanding of competitor pricing.

The simplest approach would be simply to observe the historical loss ratios in each kilometer band. Whilst simple to undertake, this analysis may be misleading for a number of reasons:

- The observed experience in each band may be driven by factors other than kilometers driven;
- There will almost certainly be insufficient exposure in each band to statistically rely on the observed experience in each band – meaning that using this approach we would over-react to bands which by random chance have had better or worse experience than expected.

[4]

(v) (a) Possible reasons include:

- Rating structure was inappropriate; potentially, this might have been due to inappropriate data, or perhaps trying to use too much granularity in the kilometers driven.

- Could be (partly) due to random claims experience – one year is a very short period.
 - There may be some selection against TGI, such as inexperienced drivers that only drive short distances choosing the policy.
 - The propensity to claim may have been different to expected.
 - Infrequent factors may play a part, such as the incidence or lack of storms, other weather events or other natural catastrophes.
 - Other underlying factors such as general weather conditions, road safety, etc.
- [2]

(b) Possible responses include:

- Changing rates to reflect claims experience. This will likely blend prior expectations (the previous rating structure) with the recent claims experience. The recent experience may be too volatile to be the only input to setting the rating structure.
- Change the kilometer bands used in the rating structure. This may involve using more or less granularity.

[2]

(c) The possible risks include:

- Changes to premiums may affect renewal rates, or for those drivers where premium rates decrease, TGI might attract “worse” risks.
- TGI may be overreacting. There may have been a short term phenomenon which affected claims experience immediately which will not continue in the long term.
- In the rating structure, there may be other variables (non-kilometer based, such as theft) which were contributing to the experience which TGI is not capturing and rating on, which other insurers are.
- The implied change to TGI’s rating structure may suggest very large changes in premium, which could unsettle the existing customer base.
- If the level of subdivision of the portfolio changes, this could potentially lead to worse profitability.
- There may be operational/execution risks if new rating variables are introduced. Can the IT systems cope? Will staff/brokers use the new schedules correctly? Incorrect premiums might be charged.
- Expense rates may be impacted if business volumes fall.

[2]

[20 Marks]

Solution 2:

(i) The information needed to determine a sound premium rate include:

Exposure –

- type of appliances sold
- no. of units sold for each product
- prices of the various types of appliances

Claim experience –

- Proportion of customers who take up the extended warranty
- Claim incident rates by type of appliances
- Cost of average repair by type of appliances
- Proportion of repairs that would not be covered by extended warranty, i.e. repairs as a result of incorrect usage or incorrect installation

Expenses –

- Initial set up costs
- Commission payable to Aria
- Ongoing management costs
- Interest expected to be earned
- Profit requirements (or capital requirements and return required on this capital)

[3]

(ii) A simple formula you could use to calculate the sound premium rate is:

premium rate = claim frequency * average claim size * (1+ expense rate + profit margin)

A more complex formula is:

premium rate = claim frequency * average claim size * $\sum$ [proportion paid in month i from policy inception * (1 + future annual inflation rate)^{i/12} / (1 + discount rate)^{i/12}] * (1+ expense rate + profit margin)

Definitions of the components:

- Claim frequency – calculated as no. of units sold * proportion taking up extended warranty * incident rate for the 3 year period
- Average claim size – average cost per repair excluding those repairs not covered. Note that frequency and size may need to be calculated by type of appliances
- Expense rate – initial set up cost and ongoing cost including commission, sales, underwriting, claims management, support/overhead, all expressed as a percentage of gross premium.
- Profit margin is what the company requires to meet their profit target on the product.
- Discount rate is the annual return expected by investing in risk free securities.

[4]

(iii) The suitability of each proposed rating factor is:

- type of product (eg. washing machine, fridge, dryer, etc.) – yes, both incident rate and repair cost would vary by the type of product
- colour – difficult to define and categorise so it would not make a good rating factor.
- energy star rating – unlikely to have an impact on cost of repair
- price of the product – yes, a good indication for the cost of repair
- model - faults generally do vary by model, but difficult to use as a rating factor given the large number of models and with new models coming out

- location sold – location would have an impact on the cost of transport to/from repair centre, but not practical to vary insurance cost by location

[3]

(iv) The differences between the two options for GHI are:

Option 1

- Effectively self insurance for Aria Appliances, and the company is acting as a reinsurer.
- The relationship is between Aria Appliances and the company.
- Administration may be simpler as Aria is dealing directly with the customers, lower administration cost.
- Depending on the reinsurance arrangement, Aria may be able to make its own underwriting and claims management decisions. This will result in higher admin expenses.
- All customers of Aria are insured.
- Aria Appliances will pay a premium to the company for reinsurance cover. How will they fund this premium? Will it be through a margin on the retail price or will the cost be borne by Aria Appliances as a squeeze on their margins. Note in Option 2 the customer will pay a premium as an explicit charge

Option 2

- Direct relationship between the customers and the company. Administration is more complex than Option 1.
- The company has full control of all underwriting and claims management decisions.
- Possible adverse selection as not all customers will take up the extended warranty product – take-up rate may be higher for those appliances more likely to generate a claim.

[3]

(v) The proposed profit share arrangement has the following problems:

- Single premium product covering a 5 year period. Should use earned premium for the year rather than received premium.
- Claim pattern not uniformly distributed over the 5 year period. Need to use an incurred pattern to allow for the 5 year policy period.
- No explicit allowance for expenses in the formula.
- The appropriateness of 20% depends on the implicit profit margin allowed in the premium
- If claims paid exceed premium received then does Aria have to pay GHI? How easy is this to enforce?

[3]

[16 Marks]

Solution 3:

- (i) The procedure to be followed for reinsurance arrangements are:
- a. The reinsurance programme should be guided by the following objectives to:
 - Maximise retention within the country
 - Develop adequate capacity
 - Secure the best possible protection for the reinsurance costs incurred
 - Simplify the administration of business
 - b. Every insurer shall maintain the maximum possible retention commensurate with its financial strength and volumes of business
 - c. Every insurer shall cede such percentage of the sum assured on each policy for different classes of insurance written in India to the Indian reinsurer as may be specified by the Authority in accordance with the provisions of part IVA of the Insurance Act 1938
 - d. The reinsurance programme of every insurer should commence from the beginning of every financial year and every insurer shall submit to the Authority, his reinsurance programmes for the coming year, 45 days before commencement of the financial year
 - e. Within 30 days of commencement of the financial year, the insurer shall file with the Authority a copy of every reinsurance treaty slip and excess of loss cover covernote in respect of that year together with the list of reinsurers and their shares in reinsurance arrangements
 - f. Insurers shall place their reinsurance business outside India with only those reinsurers who have over the last five years, counting from the year preceding for which business has to be placed, enjoyed a rating of at least BBB with Standard & Poor or equivalent rating of any other equivalent rating agency. Placements with other insurers will require approval from the Authority
 - g. Surplus, over and above the domestic reinsurance arrangements can be placed by the insurer independently with any of the reinsurers complying with sub-regulation (7) subject to a limit of 10% of the total reinsurance premium ceded outside India, being placed with any one insurer.
 - h. The Indian reinsurer shall organise domestic pools for reinsurance surpluses in Fire, Marine Hull and other classes in consultation with all insurers on basis, limits and terms which are fair and assist with maintaining the retention of business within India as close to the level achieved for the year 1999-2000.
 - i. Surplus over and above the domestic reinsurance arrangements class wise can be placed by the insurer independently with any of the reinsurers complying with sub-regulation (7) subject to a limit of 10% of the total reinsurance premium ceded outside India being placed with any one reinsurer.

- j. Every insurer shall offer an opportunity to other Indian insurers including the Indian reinsurer to participate in its Facultative and Treaty surpluses before placement of such cessions outside India
- k. The Indian Reinsurer shall retrocede at least 50% of the obligatory cessions received by it to the ceding insurers after protecting the portfolio by suitable excess of loss covers. Such retrocession shall be at original terms plus an over-riding commission to the Indian reinsurer not exceeding 2.5%. The retrocession to each ceding insurer shall be in proportion to its cessions to the Indian reinsurer.
- l. Every insurer shall be required to submit to the Authority statistics relating to its reinsurance transactions in such forms as the Authority may specify, together with its annual accounts.

[8]

- (ii) In addition to the standard points of comparison of QS and XL reinsurance, include the following points:

QS

- Pay over more premium – reduce net-to-gross ratio.
- Not as effective at reducing volatility from cat events, less than cession % due to gross event limit (losses > INR 10M but less than INR 50M will stay 100% with cedant).
- Reduces Premium Liability and OSC capital requirements in proportion to cession.
- Less than full reduction to maximum event retention (MER) capital due to event limit.
- Impact on net loss ratio depends if exchange commission > (expenses + profit).
- Transfers underwriting risk and pricing risk.

Underlying cat XL

- More effective at reducing earnings volatility from cat events.
- Net loss ratio will be increased if no covered cat. events and decreased if cats.
- Will decrease MER capital.
- Little impact on Premium Liability and OSC capital requirements.

[6]

[14 Marks]

Solution 4:

(i) The main benefits of capital allocation are to: (*standard book work*)

- ensure premiums cover the cost of capital
- help with risk selection
- identify high performing / profit generating segments
- assist with performance measurement and benchmarking
- attribute performance to assist in remuneration
- assist with strategic decision making specially with respect to venturing into newer lines, M&A etc. by assessing marginal return on capital that the new unit could generate to the existing business

Important characteristics of a good capital allocation method:

- consistent with the modelled capital for the whole business
- reflective of the relative risk and that of the business as a whole
- reflective of the interrelationships between risks and different business segments
- justifiable and easy to communicate

a) List out at least six measures of risk that may be used in a capital allocation method.

[5]

ii) Risk measures that could be used to facilitate capital allocation include:

- Standard deviation
- Variance
- Semi-variance
- Value at risk (VaR)
- Tail VaR (T-VaR)
- Excess Tail Var (X T-VaR)
- Expected Policyholder Default (EPD)
- Mean of transformed loss

[3]

iii) The desirable characteristics of a risk measure or statistic that is used to allocate capital include:

- easy to communicate and apply
- observable, measurable and not prone to manipulation
- is consistent with management's perception of risk
- quantifies risk in a manner that is understood by the decision maker
- consistent with the basis used to establish the aggregate capital requirement
- results in an internally consistent allocation of diversification benefits;
- results in a coherent allocation (with following characteristics):
 - Monotonicity
 - Linear homogeneity
 - Sub-additivity
 - Translational invariance

[4]

iv) Assumptions:

1. The current year claims and premiums are the highest (since regulations require to take average)
2. The Net and Gross claim ratios are the same to estimate GIC and NIC

Amounts in INR Crore

LoB	GWP	GIC	NWP	NIC	RSM1	RSM2	RSM
Motor	300	210	240	168	48	50.4	50.4
Health	200	130	170	110.5	34	33.15	34
Fire	100	60	90	54	18	16.2	18
						RSM	102.4
						Ratio	1.5
						MCR	153.6

[3]

v) Estimation of lognormal parameters based on the given mean and standard deviation:

$$\sigma = (\log_e((\text{standard deviation} / \text{mean})^2 + 1))^{0.5}$$

$$\mu = \log_e(\text{mean}) - \sigma^2/2$$

From the above formula:

Line of Business	Lognormal Dist.		Lognormal Parameters	
	Mean	Std Dev	μ	σ
Motor	150	20.0	5.002	0.133
Health	100	10.0	4.600	0.100
Fire	50	15.0	3.869	0.294

Capital allocation on each of the given risk measures:

Capital is set aside for that amount over and above the mean value.

VaR_{99%}If $X \sim \text{LogN}(\mu, \sigma^2)$ then $\text{Log}X \sim N(\mu, \sigma^2)$

$$\text{VaR}_{99\%}(Y \equiv \text{Log}X \sim N(\mu, \sigma^2)) = \mu + \Phi^{-1}(99\%) * \sigma$$

From the *Tables*, $\Phi^{-1}(99\%) = 2.326$. Therefore VaR values (from mu and sigma values calculated above) are:

Amts INR Crore

Line of Business	VaR _{99%}	VaR 99%	Mean	Capital to be set aside
	Normal	LogNormal		
Motor	5.3106	202.48	150.00	52.48
Health	4.8323	125.49	100.00	25.49
Fire	4.5519	94.81	50.00	44.81
			Total	122.78

Standard deviation – 2.5 times the mean

The capital to be allocated is 2.5 times the standard deviation. Therefore, allocated capital using this risk measure is:

Line of Business	mean + 2.5*sd	Mean	Capital to be set aside
	LogNormal		
Motor	200.00	150.00	50.00
Health	125.00	100.00	25.00
Fire	87.50	50.00	37.50
Total			112.50

Tail VaR

TVaR_α for $Y \sim N(\mu, \sigma^2)$ is given by $\mu + \sigma * \varphi(\beta) / (1 - \alpha)$ where φ is the probability density function of a standard normal distribution and $\beta = (VaR_{\alpha} - \mu)/\sigma$

PDF of standard normal is $f(x) = \exp(-x^2/2)/\sqrt{2*\pi}$

$$\sqrt{2*\pi} = 2.5066$$

$x = (VaR_{\alpha} - \mu)/\sigma$ will be the same for all the three lines of business since it is a fixed deviation from the mean – which is 2.3263

Therefore $\varphi(\beta)$ for all three lines = $\exp((-2.3263^2)/2)/2.5066 = 0.066807 / 2.5066 = 0.02665$

Line of Business	VaR99%		TVaR 99%	TVaR 99%	Mean	Capital to be set aside
	Normal	$\phi(\beta)$	Normal	Lognormal		
Motor	5.3106	0.02665	5.36	211.80	150.00	61.80
Health	4.8323	0.02665	4.87	129.81	100.00	29.81
Fire	4.5519	0.02665	4.65	104.72	50.00	54.72
Total						146.33

Comparison:

Amounts in INR Crore				
Line of Business	RSM requirements	VaR @ 99%	T VaR @ 99%	2.5 times SD
Motor	75.6	52.5	61.8	50.0
Health	51.0	25.5	29.8	25.0
Fire	27.0	44.8	54.7	37.5
Total	153.6	122.8	146.3	112.5

Observations:

- Capital requirements highest as per regulatory solvency requirements. Understandable in the interest of policyholders from view point of a developing economy.
- Adding across the lines for total estimate would not take into account diversification benefits. However, if the lines of business are expected to be uncorrelated then diversification benefit might not arise
- The capital allocated based on the risk measures appear aligned with the standard deviation of losses whereas the regulatory margin appears aligned with the expectation of losses (mean loss ratio) vis-a-vis business exposure.

[15]

vi) Advantages and Disadvantages:Standard deviation:Advantages:

- These are widely understood statistical measures of variability
- Easy to estimate
- Reliable results specially in case of symmetric distributions

Disadvantages:

- May give distorted results if distribution is asymmetric or if one sided risk is the concern. Semi-variance can address this aspect to a great extent
- Standard deviation does not always satisfy the monotonicity criteria of coherence – i.e. higher the risk lesser the worth of the portfolio or conversely a portfolio that has a higher worth than the other is less risky than the other.
- With skewness and kurtosis, the measure becomes more coherent but could lose the simplicity and communicability of the single risk measure

Value at RiskAdvantages:

- The VaR provides a point measure of the downside potential for a given risk, it has more practical usage
- Based on VaR for overall portfolio, it is possible to allocate capital to individual portfolios using the corresponding risk measures
- It is especially useful in assessing capital from a view point of insolvency risk as it helps estimate the threshold of insolvency at a given level of probability
- It is well understood and hence easy to communicate

Disadvantages:

- VaR does not estimate the quantum of loss beyond the threshold.
- The results may not be tenable in case of positively skewed distributions like insurance losses and thus could fail to satisfy the sub-additivity requirement of coherence i.e. combined VaR may be higher than the sum of the individual VaRs.

Tail Value at Risk**Advantages:**

- Since TVaR estimates the conditional mean of the loss distribution above the selected threshold, it is better than VaR in that it also gives the expected loss amount in case of a worst case or ruin scenario
- T VaR is a coherent risk measure as it satisfies the coherence requirements largely

Disadvantages:

- T VaR does not look at moments higher than mean – it could be of particular interest especially in case of rare but high intensity events
- T VaR, is dependent on the loss distribution above the threshold which might not always be a straightforward way to evaluate

Suggestions for improvement:

- The simple additive method of capital allocation may not provide reliable results from view point of sub additivity criterion. Can look at methods like proportional, marginal last in method, game theory (Shapley's method) and so on
- It is important to also look at correlations between lines of business while assessing capital requirements for a line – need to be more advanced by drawing a variance – covariance matrix for the lines of business to understand dependency
- Since insurance liabilities can be paid out in different periods for different lines, it is essential to look at capital requirements on a multi-period basis than only for a given point in time.

[8]

[38 Marks]

Solution 5:**i) Inferences:**

- a. Loss ratio on Gross Basis is 111% whereas on net Basis it is 87% - this could mean the insurer has a favourable composition of non-proportional arrangements that was able to contain loss ratios on a net basis.
- b. The above could also possibly mean that the insurer was able to take advantage of soft prices in the reinsurance market with low retentions and even probably a large event that led to more claims being ceded to the treaties.
- c. Retention on earned basis is 51% and that on written basis is 58% - could indicate a significant change in reinsurance arrangements / business mix from the previous year – more retail lines written leading to higher retention or decrease in obligatory ceding

- d. The ceding commission received is equal to commission paid @ 10%. This could mean either the insurer has more facultative arrangements that are typically characterized by lesser ceding commissions or that because of the soft prices in reinsurance market the ceding commissions were lower. This could be inferred as generally ceding commission rates under treaty arrangements are higher than gross commission rates.
- e. The Company's management expenses ratio is 17% which could be seen as efficient management of operating expenses in light of Indian insurance market.
- f. The Combined Ratio (NIC/NEP + Gross expenses/GWP) at 113% indicates that much of the adverse performance could be contributed by loss experience than possibly by expenses and commission.
- g. Despite the high CoR the insurer does not seem to have set aside premium deficiency reserves – this could be due to longer term policies leading to absorption of acquisition costs upfront thus leading to higher amounts upfront.
- h. There seems to be significant gap between written and earned premiums – this could indicate a significant growth witnessed in current year or possibly writing longer term policies leading to lesser amounts being translated to earnings.
- i. The above point may be substantiated by NUPR of INR 225 Crore being 65% of written premiums.
- j. The movement in NUPR of 120 Crore and Net UPR of 225 Crore as at the end of the year indicates that the business appears to have grown substantially in the current year. Of course, it could also indicate that the NUPR was lower in previous year (INR 105 Crore) owing to lesser retentions from RI arrangements
- k. The Net IBNR to NEP ratio of 13% could indicate exposure in retail lines and hence the relatively lower IBNR component. However looking at the Net UPR movement from previous year of 115% (120 / 105) vis-a-vis IBNR movement of 50% (30 / 60) the latter does not seem to have moved in line with business growth. This could indicate either change in reserving methodology in current financial year or possible reserve releases in current year owing to over-reserving in previous years.
- l. The above could also indicate significant long-term business that is leading to accretion to UPR without actually translating to expiry of risks.
- m. Investment income realised could be under 7.63% assuming the investible assets have grown over the previous year. Apparently the returns seem lower as compared with Indian market conditions in the year under observation.
- n. Net o/s claim amounts are more than Net paid amounts which is typically not the case in case of retail exposures – could indicate slower settlement process. Or a case of a major event in the current year leading to the insurer slowing its claim settlements from view point of liquidity management.
- o. P&L debit of INR 80 Crore vis-a-vis current year loss of INR 60 Crore could indicate a particularly bad year experienced by the insurer
- p. Available investments net of policyholder liabilities is INR 160 Crore and assuming these would be on a book value in line with IRDA regulations, it would be difficult to assess solvency position. Besides, IRDA solvency regulations require that the RSM is estimated at a line of business level and not at an overall level. However, at an overall level, assuming NWP and NIC are within RI factors of IRDA, the RSM is INR 105 Crore thus indicating a solvency ratio of 1.52.

- q. This requires further scrutiny particularly if market value is less than book value which could lead to ASM lower than that estimated above and the RI factors of IRDA at LoB level could be possibly higher than the retention percentages of the insurer thus warranting a higher RSM than indicated above. The effect could mean a possible breach in solvency ratio.

[8]

ii)

- a. P&L by line of business and product as applicable
- b. Classification by direct, reinsurance and coinsurance
- c. Classification by inward reinsurance and ceded reinsurance
- d. Business bifurcation by term (annual and otherwise)
- e. Business bifurcation by persistency (new vs. Renewal)
- f. Segregation of loss amounts by attritional and CAT losses
- g. Solvency position by line of business
- h. Invested assets allocation into shareholder and policy holder funds
- i. Policy level information to analyze trends in premium vis-a-vis losses viz., average premiums, average sum insured and so on
- j. Cash flow / fund flow statement

[4]

[12 Marks]
